

Francesco Cavina

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Quantitative data and AI engineer with a BSc in Econometrics and Data Science from Vrije Universiteit Amsterdam and incoming MSc studies in Econometrics and Operations Research and Business Analytics. Experience in operational trading, Python and SQL workflows, automated data pipelines, dashboards, model evaluation and self service data tools. Strong fit for finance operations work that combines accurate data handling, cross team information flows, process improvement and practical analysis in a trading environment.

TECHNICAL AND ANALYTICAL SKILLS

- **Quantitative analysis:** econometrics, probability, statistics, optimisation, forecasting, model evaluation, walk forward validation
- **Finance and operations support:** trading workflows, recurring forecasts, dashboarding, performance visualisation, assumption testing, documentation
- **Programming and data:** Python, SQL, PySpark, data ingestion, automated workflows, schema inference, data quality workflows
- **Automation and AI tools:** LLM workflows, RAG, ChromaDB, OpenAI API, Hugging Face, local LLM workflows, structured prompts
- **Engineering tools:** Git, Docker, FastAPI, Azure, Databricks, Airflow, Kubernetes, Terraform
- **Communication:** teaching, technical explanation, user focused prototypes, translating model output into practical recommendations

RELEVANT EXPERIENCE

Data & AI Platform Engineer

Robodata
Netherlands

September 2025 - June 2026, Amsterdam,

- Developed automated Bronze, Silver and Gold data pipelines in Python, PySpark and SQL.
- Built metadata driven ingestion, schema inference and PII classification workflows for enterprise data platforms.
- Built local LLM workflows for data discovery, field classification and metadata generation.
- Created self service tooling and data interfaces so non specialist users could query, understand and reuse internal datasets.
- Documented workflows, platform components and design choices across Docker, Kubernetes, Terraform, Azure, Databricks, Airflow and SQLAlchemy.

Operational Trader

QuantFi
Netherlands

October 2022 - May 2023, Amsterdam,

- Developed and tested Python based trading strategies using Random Forest, XGBoost, neural networks and walk forward validation.
- Built Python and SQL workflows for data ingestion, retraining and daily forecasts used in recurring trading decisions.
- Created dashboards comparing model signals, realised performance and forecast quality.
- Evaluated outputs for overfitting, changing market conditions and practical trading use, in a setting where speed and disciplined decisions mattered.

Teaching Assistant

Vrije Universiteit Amsterdam
Netherlands

June 2022 - November 2024, Amsterdam,

- Taught weekly Probability Theory tutorials to first year Econometrics students.
- Explained abstract mathematical concepts clearly, with focus on reasoning, assumptions and formal problem solving.
- Built a student portfolio system and represented the EOR programme during Open Days.

EDUCATION

MSc in Econometrics and Operations Research

Track Business Engineering and Data Driven Decision Making • Vrije Universiteit Amsterdam
• Amsterdam, Netherlands • 2026-2029

MSc in Business Analytics

Track Computational Intelligence • Vrije Universiteit Amsterdam
• Amsterdam, Netherlands • 2026-2029

Bachelor of Science in Econometrics and Data Science

Vrije Universiteit Amsterdam • Amsterdam, Netherlands • 2021-2024 • 7.8/10 •
Thesis: Nonstandard Probabilities in an Internal Probability Space, grade 8.5/10

Minor in Pure Mathematics

University of Leeds • Leeds, England
• 2024 • 8.5/10

PROJECTS

- 1) [AI Powered Personal Website](https://github.com/FrancescoCavina02/My-Website) • <https://github.com/FrancescoCavina02/My-Website> • November 2025 - present
Next.js, TypeScript, FastAPI, Python, Tailwind CSS
Built a full stack portfolio with a FastAPI backend that parses an Obsidian vault into structured API endpoints for notes, quotes and chat functionality. Designed it as a self service knowledge interface, combining content modelling, backend automation, frontend usability and LLM based interaction.
- 2) [RAG Knowledge Chatbot](https://github.com/FrancescoCavina02/Spiritual-chatbot) • <https://github.com/FrancescoCavina02/Spiritual-chatbot> • November 2025 - present
Python, FastAPI, ChromaDB, sentence transformers, LLM APIs
Built a retrieval augmented chatbot using semantic chunking, vector search, structured prompts and citation based responses. Focused on reliable retrieval, traceable answers, hallucination reduction and reusable workflow design.
- 3) [Dodgeball Club Amsterdam](https://github.com/FrancescoCavina02/DCA-website) • <https://github.com/FrancescoCavina02/DCA-website> • June 2025 - present
Built and maintain a public website using HTML, CSS and JavaScript. Focused on clear navigation, simple content management and practical usability for club members.

Podcasts

[Podcast Link](#) - Created two independent interview podcasts with researchers in physics, cosmology, mathematics and philosophy.